

Example 8 *Driving theory tests: which significance level?*

Example 5 considered testing the hypotheses

$$H_0 : \mu = 51.6\%, \quad H_1 : \mu < 51.6\%,$$

where μ is the mean driving theory test pass rate nationally over the period April 2014–March 2015. The observed value of the test statistic was $z \simeq -1.693$. Using a 5% significance level, the critical value for this one-sided test was $c_1 = -1.645$, which led us to reject H_0 since $-1.693 < -1.645$.

However, if a 1% significance level had been used instead, then the critical value would have been

$$c_1 = q_{0.01} = -q_{0.99} = -2.326,$$

which would mean that H_0 would *not* have been rejected, since $-2.326 < -1.693$.

Both the 1% and 5% critical values, together with the observed value of the test statistic, are marked on a plot of the null distribution in Figure 6.

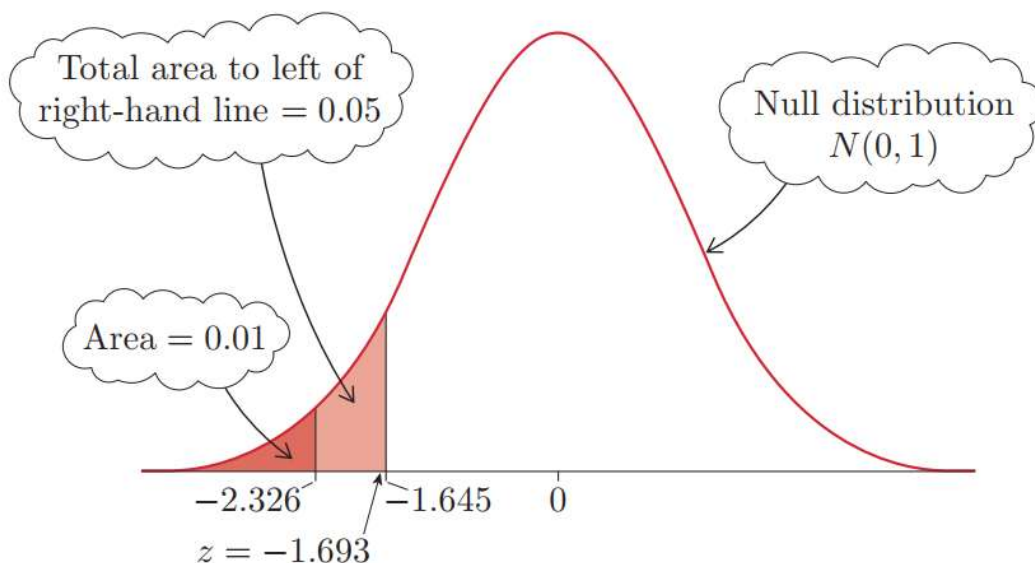


Figure 6 Null distribution $N(0, 1)$ with the 1% critical value -2.326 , the 5% critical value -1.645 , and the observed value $z = -1.693$ marked

In this section, a different approach to testing hypotheses is considered in which a significance level does not need to be specified for a test. Instead of leading to a stated decision such as ‘reject H_0 ’, this alternative approach

to testing hypotheses leads to a number called the **significance probability** which, loosely speaking, describes the extent to which the data provide evidence against the null hypothesis. The general idea is that the lower the significance probability, the more evidence the data provide against the null hypothesis. The significance probability is denoted by p ; it is because of this that it is often referred to as the **p -value**.

The use of significance probabilities has become the most common approach for testing hypotheses and, unless you are specifically asked to use critical values and a specific significance level, this is the approach that you should use to perform hypothesis tests.

The idea of a significance probability, or p -value, is to describe the extent to which the data provide evidence against the null hypothesis, rather than to decide whether or not there is sufficient evidence to reject the null hypothesis. This is done by calculating the probability of obtaining a value of the test statistic that is 'at least as extreme as' the observed value of the test statistic when the null hypothesis is true. This is intended as a measure of how likely we are to have observed the data we have observed, or something even more extreme, if the null hypothesis is true. Such a probability, the p -value, will be low when the data are surprising under H_0 , so a low p -value corresponds to evidence against the null hypothesis. This is illustrated in Example 9.

What is a p -value?

In statistics, a p -value (probability value) is a number between 0 and 1 that measures **how likely your results are to happen by pure chance**.

When scientists run an experiment, they start by assuming nothing special is happening (this is called the **null hypothesis**). The p -value answers the question:

"If nothing special is actually happening, what is the probability of seeing results this extreme just by luck?"

A Simple Example: A Coin Flip

Imagine your friend gives you a regular coin and says it is completely fair.

- You flip the coin **10 times**, and it lands on **Heads all 10 times**.
- The probability of getting 10 heads in a row with a fair coin by pure luck is:

$$\left(\frac{1}{2}\right)^{10} = \frac{1}{1,024} \approx 0.001 \text{ (or 0.1\%)}$$

Here, your p -value ≈ 0.001 . Because this number is so tiny, it is very unlikely to happen by pure luck. You would conclude that the coin is probably **not** fair!

How to Interpret a p -value

Researchers usually compare the p -value to a threshold called the **significance level** (often denoted by the Greek letter alpha, α), which is usually set at 0.05 (5%):

1. **Small p -value** ($p \leq 0.05$):

- The result is very unlikely to happen just by random chance.
- We say the result is **statistically significant**, meaning something real is likely going on.

2. **Large p -value** ($p > 0.05$):

- The result could easily happen just by random chance.
- There is not enough evidence to claim anything unusual is happening.

Summary Rule of Thumb

- **Low p -value** $\rightarrow$ **Surprise!** It's too weird to be pure chance.
- **High p -value** $\rightarrow$ **Normal.** It could easily just be luck.

Example 9 *Driving theory tests: p -value*

Continuing on from Example 8, the observed value of the test statistic is $z = -1.693$. This is the test statistic for the one-sided test with hypotheses

$$H_0 : \mu = 51.6\%, \quad H_1 : \mu < 51.6\%,$$

which means that values in the lower tail only will provide evidence against H_0 . So any value of z less than -1.693 will be considered to be more extreme than the observed value of the test statistic. Then, since the p -value is the probability of obtaining a value of the test statistic that is 'at least as extreme as' the observed value of the test statistic when the null hypothesis is true, this means that

$$p = P(Z \leq -1.693),$$

where $Z \sim N(0, 1)$ when the null hypothesis is true. So

$$\begin{aligned} p &= P(Z \leq -1.693) = P(Z \geq 1.693) \\ &= 1 - P(Z < 1.693) \simeq 1 - P(Z < 1.69) \\ &= 1 - \Phi(1.69) = 1 - 0.9545 = 0.0455. \end{aligned}$$

(The value -1.693 was rounded to -1.69 to facilitate use of the table of standard normal probabilities used in M248.) The p -value is illustrated in Figure 7 (overleaf).

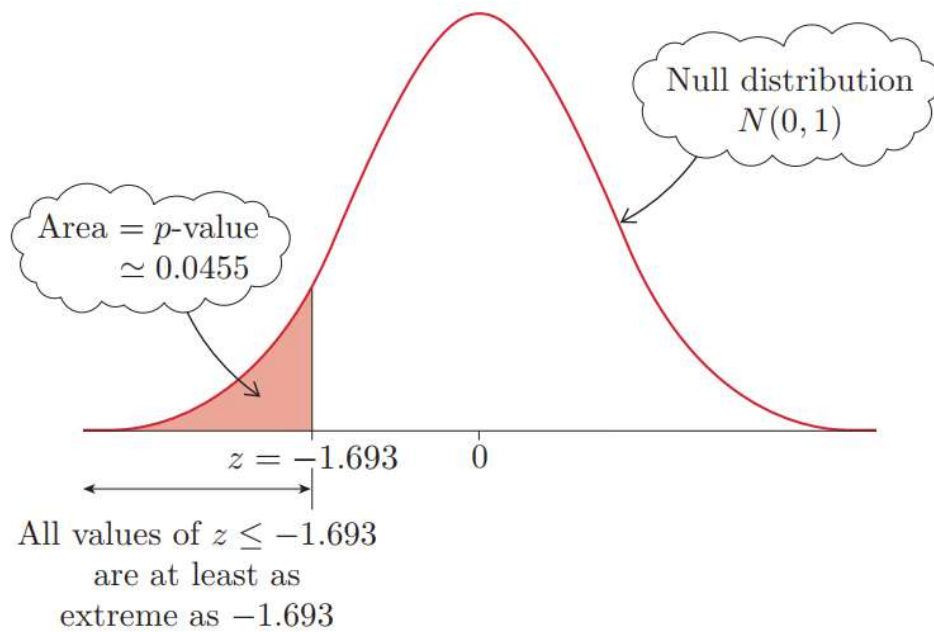


Figure 7 Null distribution $N(0, 1)$ with the p -value for the observed value $z = -1.693$ shown

So now that we have calculated a p -value, what do we do with it? What does it tell us about the evidence against the null hypothesis? Well, there aren't any hard and fast rules as to how to interpret p -values, but a rough guide is given in Table 3. Please do be aware that these are *only guidelines* and there will always be situations in which any rule of thumb will be found wanting!

Table 3 Interpreting p -values

p -value	Rough interpretation
$p > 0.10$	little or no evidence against H_0
$0.05 < p \leq 0.10$	weak evidence against H_0
$0.01 < p \leq 0.05$	moderate evidence against H_0
$p \leq 0.01$	strong evidence against H_0